

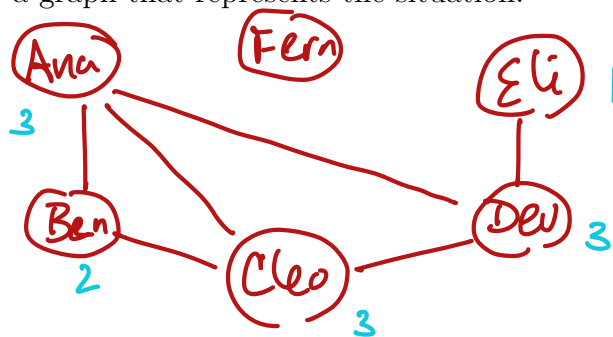
Worksheet: Modeling with Graphs

Group Names: _____

1. Six students are in a study group. The table says who has already worked together.

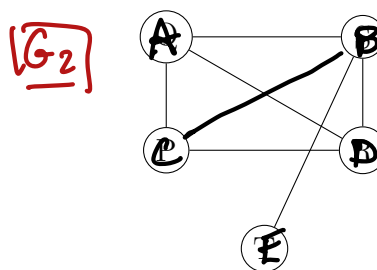
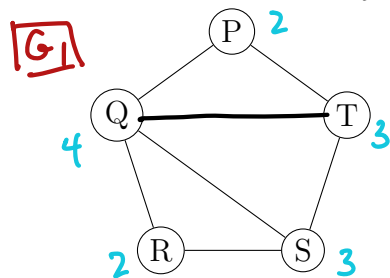
student	has worked with
Ana	Ben, Cleo, Dev
Ben	Ana, Cleo
Cleo	Ana, Ben, Dev
Dev	Ana, Cleo, Eli
Eli	Dev
Fern	—

- (a) Draw a graph that represents the situation.



- (b) What are the vertices in your model? students
- (c) What are the edges? study partners
- (d) Is your graph connected? No What does that mean **about the study group**?
Fern hasn't worked with anyone yet!

2. Below are two graphs. Two graphs are different if there is a feature that one has that the other does not.



Same graph? No Why or why not? (explain)

Vertex E is degree 1 but all vtx in G_1 have degree at least 2

3. We want to model the following situations.

- (a) A snowplow must clear every street in a neighborhood.

vertices: intersections edges: roads

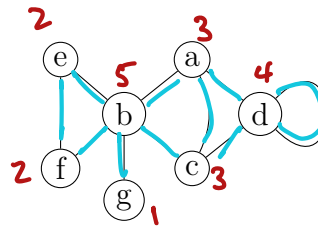
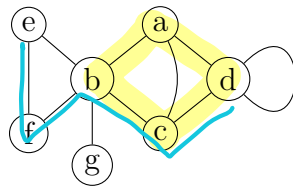
- (b) A delivery driver must stop at ~~every house on a street~~ ^{several businesses}, once each.

vertices: businesses edges: roads

- (c) A utility wants to run fiber to every village in a region as cheaply as possible.

vertices: villages edges: potential fiber connections

4. **Degree.** Graph **W** is shown below, twice.



- (a) Label each vertex of the **right-hand** copy with its degree. Remember a loop counts twice.

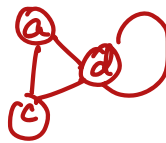
- (b) Which vertex has the largest degree? b

- (c) How many edges does **W** have? 10

- (d) Find a circuit and highlight/draw it on the left-hand graph. b a d e

- (e) Find a path from *e* to *d*. Write the sequence of vertices e f b c d

- (f) Is there a vertex you could delete that would leave the graph **not** connected? b
Why?



No path from e to g

5. Fill in the table using the graphs on this worksheet.

graph	sum of all the degrees	number of edges
study group (#1)	<u>12</u>	<u>6</u>
graph in #2, left picture ^{G₁}	<u>14</u>	<u>7</u>
graph W	<u>20</u>	<u>10</u>

Find a pattern: the sum of all the degrees of the vertices of a graph is always 2 times the number of edges.

Why should that be true? Each end of an edge is counted once at each vertex, so each edge (with 2 ends) is counted twice!